PROJECT REPORT

Speech Emotion Recognition Using Deep Learning

Under the guidance of:

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DEAL, DRDO, Dehradun

Submitted by:

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CERTIFICATE

This is to certify that **Mr. Arsh Mohan Nishant**, a Bachelor of Technology (B.Tech) student in **Computer Science and Engineering at KIIT University, Bhubaneswar**, has successfully completed the project titled **“Speech Emotion Recognition Using Deep Learning”** at **Defence Electronics Applications Laboratory (DEAL), Dehradun**, as a part of his internship.

The internship was carried out during the period from **12th May 2026 to 26th June 2026**, under the guidance and supervision of **Mr. Sandeep Kishore,** Scientist ‘E’, Artificial Intelligence & Advanced Tactical Networking and Communication Systems (AI & ATNCS), DEAL, DRDO, Dehradun, Uttarakhand.

During the internship, Mr. Arsh Mohan Nishant demonstrated dedication, sincerity, and strong technical skills in the field of Artificial Intelligence and Deep Learning. He successfully worked on the development of a Speech Emotion Recognition system involving audio preprocessing, feature extraction, deep learning model development, training, and performance evaluation.

(Sandeep Kishore)	**(Bhanu Pratap Singh)**
Scientist ‘E’	Scientist ‘G’
AI & ATNCS	Group Director, AI & TNACS
DEAL, DRDO	DEAL, DRDO

ACKNOWLEDGEMENT

I am highly obliged to **Shri Manoj Kumar Dhaka, Scientist 'H' & Outstanding Scientist, Director, Defence Electronics Applications Laboratory (DEAL), Dehradun**, for granting me permission to undertake my internship at DEAL during the period from **12th May 2026 to 26th June 2026**.

I express my sincere gratitude to **Shri Bhanu Pratap Singh, Scientist 'G', Group Director, AI & TNACS**, for his support, encouragement, and guidance throughout the course of my project work. I would also like to convey my heartfelt thanks to **Shri Sandeep Kishore, Scientist 'E', AI & ATNCS**, for his valuable guidance, continuous support, constructive suggestions, and technical mentorship during the development of my project titled **"Speech Emotion Recognition Using Deep Learning."** His guidance played a crucial role in the successful completion of this work.

I would also like to express my sincere gratitude to **Mr. Ashok Chaudhary Sir** for his encouragement, support, and assistance in facilitating this internship opportunity at DRDO-DEAL, Dehradun. His guidance and motivation helped me gain this valuable learning experience.

I am thankful to the **HRD Group** for their cooperation, assistance, and guidance during my stay at DEAL. I would also like to extend my sincere gratitude to the **Technical Information Centre (TIC)** for providing standard reference materials relevant to my project work.

Last but not the least, I sincerely thank all the scientists, engineers, and staff members of DEAL who directly or indirectly contributed their valuable time, support, and expertise towards the successful completion of this project.

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DEFENCE ELECTRONICS APPLICATIONS LABORATORY (DEAL)

Defence Electronics Applications Laboratory (DEAL), Dehradun is a premier research and development laboratory under the **Defence Research & Development Organisation (DRDO)**, Ministry of Defence, Government of India. Established initially in 1965 as the Himalayan Radio Propagation Unit (HRPU) and later relocated to Dehradun, DEAL has evolved into a major DRDO system laboratory focused on advanced defence communication and electronics technologies. Its primary mission encompasses the design, development, and integration of communication systems and data links for defence applications, including software-defined radios, satellite, and millimetre-wave communication systems and long-haul communication links, and surveillance technologies. DEAL also undertakes satellite image reception and processing, network management systems, and data fusion environments critical for modern defence operations.

DEAL's work supports key defence platforms and programmes, such as airborne early warning and control systems, unmanned aerial vehicles, and mission-critical tactical communication networks. The laboratory's contributions have enabled indigenous solutions for secure battlefield communication, robust data transmission, and real-time information exchange, reinforcing India's self-reliance in defence electronics.

DEAL – The Organization

DEAL is a major systems laboratory of Defence Research & Development Organization (DRDO) under Ministry of Defence, Govt. of India, located in Dehradun, the capital of Uttarakhand (Dev Bhoomi), pursuing state-of-the-art technologies in the areas of military wireless communications & Surveillance systems (Satellite based systems for Secure Communication and Surveillance, Software Radios, Data Links, Millimetre Wave Communication, Image Processing & Analysis techniques) leading to delivery of fully engineered systems and solutions for meeting requirements of the armed forces.

Vision

The Defence Electronics Applications Laboratory (DEAL), a key laboratory of the DRDO, envisions leading in the design and development of advanced, indigenous, and secure electronic communication, surveillance, and warfare systems. Its focus lies on fostering self-reliance in military wireless, satellite communication, and data links to ensure a technological edge for the Indian Armed Forces.

Key aspects of DEAL, DRDO's vision include:

- **Indigenous Technology Development:** Developing state-of-the-art radio communication, satellite communication, and data links.
- **Modernization & Future Trends:** Shifting towards next-generation technologies like artificial intelligence (AI), software-defined radios (SDR), and electronic scanning array radars.
- **Strategic Collaboration:** Engaging with start-ups, academia, and industry to explore futuristic communication technologies.
- **Core Research Focus:** Developing specialized areas such as microwave and millimetre-wave communication, as well as electronic warfare, to support the Indian Armed Forces.

Mission

"The mission of Defence Electronics Applications Laboratory (DEAL) is to design, develop, and deliver advanced, indigenous, and mission-critical radio communication systems, data links, and satellite communication technologies to empower the Indian Armed Forces."

ABSTRACT

This project presents the design and implementation of a Speech Emotion Recognition (SER) system based on a **Fusion Deep Learning Model** that combines **CNN, BiLSTM, and Attention** mechanisms. The system aims to automatically identify human emotions from speech signals under varying speaking styles, recording conditions, and speaker characteristics.

The proposed approach utilizes Mel-Frequency Cepstral Coefficients (**MFCC**), Delta MFCC, Delta-Delta MFCC, and **Mel Spectrogram** features to capture both temporal and spectral information from speech signals. The MFCC-based features are processed through a Bidirectional Long Short-Term Memory (BiLSTM) network with an Attention mechanism to model temporal dependencies, while Mel Spectrogram features are processed through a Convolutional Neural Network (CNN) to learn spatial representations. The extracted features are fused through a feature fusion layer and passed through fully connected layers for emotion classification.

To improve robustness and generalization, **various preprocessing** and **data augmentation** techniques are employed, including normalization, pitch shifting, time stretching, and noise injection. The system is trained and evaluated using emotional speech samples collected from the CREMA-D, TESS, and RAVDESS datasets, covering six emotion classes: **Angry, Disgust, Fear, Happy, Neutral, and Sad.**

The proposed Fusion Model provides an effective framework for automatic speech emotion recognition and demonstrates the capability of deep learning-based feature fusion techniques for emotion-aware speech analysis applications.

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INTRODUCTION

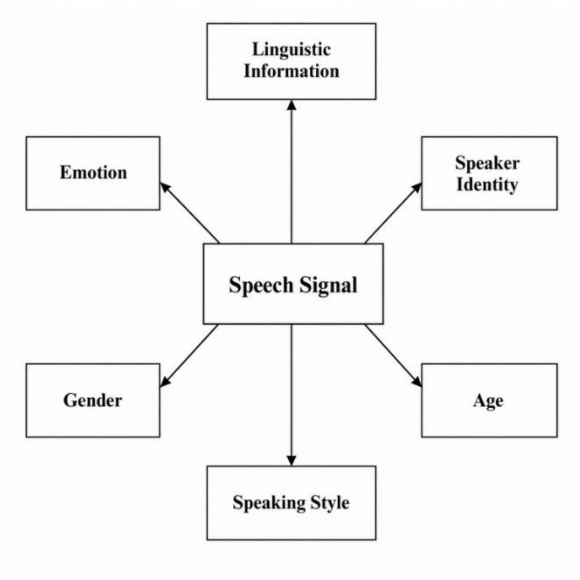
Human communication is a complex process that involves not only the exchange of information through words but also the expression of emotions, intentions, and psychological states. Among various modes of communication, speech is one of the most natural and effective mediums used by humans to convey information. In addition to linguistic content, speech carries emotional cues that reflect the speaker's feelings and state of mind, playing a crucial role in interpersonal communication.

A speech signal contains multiple levels of information beyond the spoken words. As shown in Figure 1, speech conveys linguistic information as well as speaker-specific characteristics such as emotion, speaker identity, gender, age, and speaking style. These characteristics are embedded within the acoustic properties of speech and provide valuable information about the speaker. Among them, emotion is one of the most significant factors influencing human interaction and communication.

With the rapid advancement of Artificial Intelligence (AI) and Human-Computer Interaction (HCI), there is an increasing demand for intelligent systems capable of understanding and responding to human emotions. Speech Emotion Recognition (SER) addresses this challenge by automatically identifying emotional states from speech signals. By analyzing acoustic characteristics such as pitch, energy, intensity, speaking rate, and spectral information, SER systems can classify emotions such as anger, happiness, sadness, fear, disgust, and neutrality.

In this project, a deep learning-based Speech Emotion Recognition system is developed using a feature fusion approach that combines MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features. A hybrid CNN-BiLSTM-Attention architecture is employed to capture both spectral and temporal characteristics of speech signals for emotion classification. The system is trained and evaluated using the RAVDESS, CREMA-D, and TESS datasets covering six emotion categories: Angry, Disgust, Fear, Happy, Neutral, and Sad.

The objective of this project is to develop an efficient and robust Speech Emotion Recognition system capable of automatically analyzing speech signals and identifying the underlying emotional state of a speaker. Such systems have potential applications in healthcare, education, customer service, intelligent virtual assistants, human-computer interaction, and defence communication systems.



Information Contained in a Speech Signal

1. Speech Emotion Recognition

Speech Emotion Recognition (SER) is the process of automatically identifying and classifying human emotions from speech signals. Unlike traditional speech processing systems that focus on recognizing spoken words, SER aims to determine the emotional state of a speaker by analyzing how the speech is delivered.

Human speech contains not only linguistic information but also emotional cues such as pitch, energy, intensity, speaking rate, and spectral characteristics. These acoustic properties vary with different emotions and can be used to distinguish emotional states such as Angry, Disgust, Fear, Happy, Neutral, and Sad.

With the rapid advancement of Artificial Intelligence and Deep Learning, Speech Emotion Recognition has become an important research area in Human-Computer Interaction. In this project, a deep learning-based feature fusion model is developed using MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features. These features are processed through a hybrid CNN-BiLSTM-Attention architecture for automatic emotion classification.

2. Importance of Emotion Recognition

Emotion plays an important role in human communication. During conversations, people not only understand spoken words but also interpret the emotional state of the speaker through variations in tone, pitch, intensity, and speaking style. Understanding emotions helps in improving communication and decision-making.

With the advancement of Artificial Intelligence and Human-Computer Interaction technologies, there is an increasing need for systems capable of understanding human emotions. Speech Emotion Recognition enables machines to automatically identify emotions from speech signals, making interactions more natural and effective.

Emotion recognition has applications in healthcare, customer support, education, virtual assistants, and defence communication systems. The ability to automatically recognize emotions can help develop intelligent systems that better understand human behavior and provide context-aware responses.

3. Challenges in Speech Emotion Recognition

Speech Emotion Recognition is a challenging task due to the complex nature of human speech and emotions. The same emotion may be expressed differently by different speakers depending on factors such as age, gender, accent, speaking style, and cultural background.

Background noise, recording conditions, and variations in speech quality can also affect the performance of emotion recognition systems. In addition, some emotions share similar acoustic characteristics, making them difficult to distinguish accurately.

Developing a robust Speech Emotion Recognition system therefore requires effective preprocessing, feature extraction, and deep learning techniques capable of capturing both temporal and spectral information from speech signals.

4. Classification of Emotions

Emotions represent the psychological state of an individual and are reflected through variations in speech characteristics such as pitch, energy, speaking rate, and spectral patterns. In Speech Emotion Recognition systems, emotions are categorized into predefined classes for automatic classification.

In this project, six emotional categories are considered:

- **Angry:** Characterized by high energy, increased pitch, and strong vocal intensity.
- **Disgust:** Expressed through irregular speech patterns and distinctive vocal characteristics.
- **Fear:** Often associated with rapid speech, pitch fluctuations, and vocal tension.
- **Happy:** Generally characterized by higher pitch, increased energy, and expressive speech.
- **Neutral:** Represents normal speech without significant emotional influence.
- **Sad:** Typically associated with lower energy, slower speaking rate, and reduced vocal intensity.

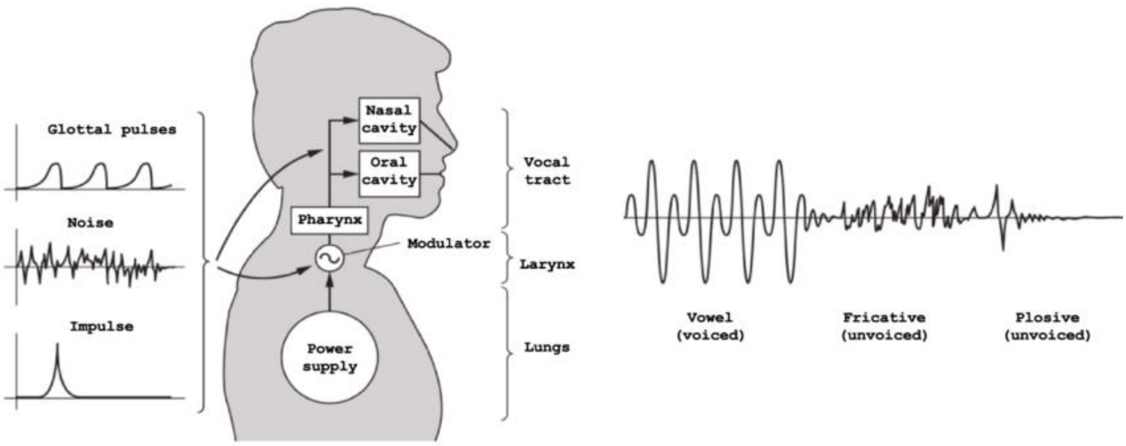
5. Human Speech Production Mechanism

Human speech is produced through a complex interaction of the respiratory system, vocal cords, and vocal tract. The lungs act as the power source by providing a continuous airflow, which passes through the larynx containing the vocal cords. The vibration of the vocal cords generates a sound source that is further shaped by different parts of the vocal tract, including the pharynx, oral cavity, and nasal cavity, to produce speech sounds.

Depending on the mode of excitation, speech signals may be generated through voiced, unvoiced, or mixed speech production mechanisms. Voiced sounds are produced by periodic vibrations of the vocal cords, while unvoiced sounds are generated by turbulent airflow without vocal cord vibration. The characteristics of the resulting speech signal are influenced by the shape and configuration of the vocal tract during articulation.

The process of speech production is directly related to emotion expression. Different emotional states affect speech characteristics such as pitch, energy, speaking rate, vocal intensity, and spectral properties. For example, angry speech often exhibits higher energy and pitch variations, whereas sad speech is generally associated with lower energy and slower speaking rates. These variations create distinct acoustic patterns that can be analyzed by Speech Emotion Recognition systems.

Understanding the mechanism of human speech production is therefore important for extracting meaningful speech features and developing accurate emotion recognition models. The acoustic information generated during speech production forms the foundation for feature extraction and emotion classification in modern Speech Emotion Recognition



systems.

Human Speech Production Mechanism and Generation of Speech Signals.

PROBLEM STATEMENT AND MOTIVATION

Speech Emotion Recognition has emerged as an important research area due to the growing demand for intelligent systems capable of understanding human emotions. The objective of Speech Emotion Recognition is to automatically identify the emotional state of a speaker from speech signals. However, achieving accurate emotion classification remains a challenging task due to the complex and dynamic nature of human speech.

The major challenges associated with Speech Emotion Recognition are:

- Variations in speech characteristics due to differences in age, gender, accent, and speaking style.

- Presence of background noise and recording artifacts that affect speech quality.

- Similar acoustic characteristics among different emotions, leading to misclassification.

- High variability in emotional expression across different speakers.

- Difficulty in capturing both temporal and spectral information using traditional approaches.

Traditional Speech Emotion Recognition systems primarily relied on handcrafted features and conventional machine learning algorithms such as Support Vector Machines (SVM) and Random Forest classifiers. Although these methods achieved reasonable performance, they often struggled to model the complex relationships present in speech signals and required extensive feature engineering.

The motivation behind this project is to develop an efficient and robust Speech Emotion Recognition system capable of overcoming these limitations. To address these challenges, a feature fusion approach is adopted by combining MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features. A hybrid CNN-BiLSTM-Attention architecture is employed to effectively learn both spectral and temporal speech representations for emotion classification.

The proposed system aims to provide an effective framework for automatic emotion recognition from speech and contribute towards the development of intelligent emotion-aware applications.

Overall Problem-to-Solution Flow of Speech Emotion Recognition System

Speech Signal

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Challenges in Emotion Recognition

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Feature Extraction

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Deep Learning Model

↓

Emotion Classification

Comparison Between Traditional and Proposed Speech Emotion Recognition Approaches

Aspect	Traditional Methods	Proposed System
Feature Extraction	Single Handcrafted Features	MFCC + Delta MFCC + Delta-Delta MFCC + Mel Spectrogram
Modelling	SVM / Random Forest	CNN + BiLSTM + Attention
Temporal Learning	Limited	Strong
Feature Representation	Individual Features	Feature Fusion
Noise Robustness	Moderate	Improved Through Augmentation
Scalability	Moderate	High

APPLICATIONS OF SPEECH EMOTION RECOGNITION

Speech Emotion Recognition (SER) has gained significant attention in recent years due to its ability to automatically identify human emotions from speech signals. By analyzing acoustic characteristics such as pitch, energy, intensity, and spectral features, SER systems can determine the emotional state of a speaker. The advancement of Artificial Intelligence and Deep Learning has further improved the effectiveness of emotion recognition systems, making them suitable for a wide range of real-world applications.

1. Healthcare Applications

Speech Emotion Recognition can assist healthcare professionals in monitoring the emotional and psychological condition of patients. Changes in emotional state often provide important indicators of stress, anxiety, depression, and other mental health conditions. SER systems can support continuous and non-invasive monitoring, enabling early detection and timely intervention.

2. Customer Support Systems

In customer service environments, Speech Emotion Recognition can be used to analyze customer emotions during conversations. By identifying emotions such as frustration, anger, or satisfaction, organizations can evaluate service quality and improve customer experience. Emotion-aware systems can also assist support agents in handling customer interactions more effectively.

3. Education and E-Learning

Speech Emotion Recognition can enhance online learning environments by analyzing student engagement and emotional responses during learning sessions. Detecting emotions such as confusion, boredom, or interest can help educators assess learning effectiveness and adapt instructional strategies accordingly. This can contribute to a more personalized and interactive learning experience.

4. Defence Applications

Speech Emotion Recognition has potential applications in defence and security systems where understanding the emotional state of personnel can improve situational awareness and communication effectiveness. Emotion analysis can support decision-making, stress monitoring, and human-machine interaction in operational environments. Such systems can contribute to the development of intelligent communication technologies capable of responding to the emotional context of users.

LITERATURE SURVEY

Speech Emotion Recognition (SER) has attracted significant research interest in the fields of speech processing, machine learning, and artificial intelligence. Early SER systems primarily relied on handcrafted features such as MFCC, pitch, and energy combined with traditional machine learning algorithms for emotion classification.

Recent advances in deep learning have significantly improved SER performance. Convolutional Neural Networks (CNNs) have been used to learn spatial representations from spectrograms, while LSTM and BiLSTM networks effectively capture temporal dependencies in speech signals. Attention mechanisms have further enhanced emotion recognition by focusing on important speech segments.

Several studies have also explored feature fusion techniques by combining MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features. These approaches have demonstrated improved performance by capturing complementary spectral and temporal information. Based on these findings, the present work adopts a feature fusion strategy combined with a CNN-BiLSTM-Attention architecture for Speech Emotion Recognition.

Summary of Existing Speech Emotion Recognition Studies

Study	Features Used	Model	Key Finding
Traditional SER Systems	MFCC, Pitch, Energy	SVM, HMM	Moderate performance
CNN-based SER	Mel Spectrogram	CNN	Better spatial feature learning
LSTM-based SER	MFCC	LSTM	Improved temporal modeling
Attention-based SER	MFCC + Spectrogram	BiLSTM + Attention	Better contextual understanding
Feature Fusion SER	MFCC + Delta + Mel	Hybrid Networks	Improved classification accuracy

SYSTEM OVERVIEW

The proposed Speech Emotion Recognition (SER) system is designed to automatically identify emotions from speech signals using a deep learning-based feature fusion approach. The system processes speech recordings through multiple stages, including audio preprocessing, feature extraction, feature fusion, deep learning-based classification, and emotion prediction. The overall workflow of the proposed system is shown in Figure 5.

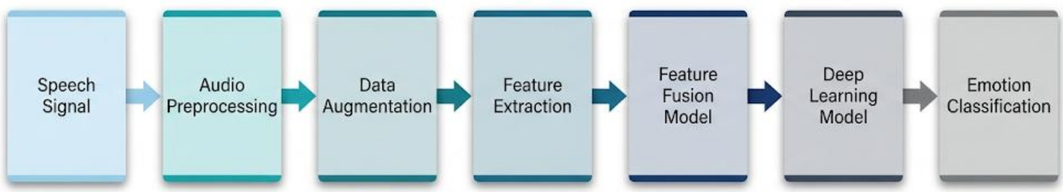
Initially, speech signals are collected from emotional speech datasets and subjected to preprocessing operations such as resampling, normalization, and fixed-length audio generation. These steps ensure consistency in the input data and improve the quality of extracted features.

After preprocessing, multiple acoustic features are extracted from the speech signals. The extracted features include Mel-Frequency Cepstral Coefficients (MFCC), Delta MFCC, Delta-Delta MFCC, and Mel Spectrograms. These features capture important spectral and temporal characteristics associated with different emotional states.

The extracted features are then combined using a feature fusion strategy. MFCC, Delta MFCC, and Delta-Delta MFCC features are processed through a BiLSTM network with an Attention mechanism to capture temporal dependencies in speech signals. Simultaneously, Mel Spectrogram features are processed through a Convolutional Neural Network (CNN) to learn spatial representations. The learned representations are fused to obtain a comprehensive feature representation.

Finally, the fused features are passed through fully connected layers for emotion classification. The proposed system classifies speech samples into six emotional categories: Angry, Disgust, Fear, Happy, Neutral, and Sad. This hybrid CNN-BiLSTM-Attention framework enables effective learning of both spectral and temporal information, thereby improving emotion recognition performance.

Overall Workflow of the Proposed Speech Emotion Recognition System



DATASET DESCRIPTION

The performance of a Speech Emotion Recognition (SER) system largely depends on the quality and diversity of the training data. To ensure robust emotion classification, three publicly available emotional speech datasets, namely CREMA-D, TESS, and RAVDESS, were used in this work. These datasets contain speech recordings from multiple speakers expressing different emotions under varying recording conditions.

The selected datasets were combined to create a unified emotional speech corpus covering six emotion categories: Angry, Disgust, Fear, Happy, Neutral, and Sad. The combined dataset provides diversity in terms of speaker characteristics, speaking styles, gender, and recording environments, thereby improving the generalization capability of the proposed model.

1. CREMA-D Dataset

The Crowd-Sourced Emotional Multimodal Actors Dataset (CREMA-D) is a widely used emotional speech dataset containing recordings from multiple actors expressing different emotions. The dataset includes speech samples recorded by speakers of different ages, genders, and ethnic backgrounds, making it suitable for Speech Emotion Recognition research.

Number of Samples: **7,442**

2. TESS Dataset

The Toronto Emotional Speech Set (TESS) contains speech recordings spoken by female speakers expressing various emotions. The dataset provides clear and high-quality emotional speech samples and is commonly used for evaluating emotion recognition systems.

Number of Samples: **2,400**

3. RAVDESS Dataset

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is a benchmark emotional speech dataset containing professionally recorded emotional utterances. In this work, only the speech recordings were utilized for emotion recognition.

Number of Samples: **1,056**

4. Combined Dataset

The three datasets were merged into a single dataset to increase data diversity and improve model robustness. After combining the datasets and selecting the common emotion classes, a total of **10,898 speech samples** were obtained.

Dataset Distribution

Dataset	*Number of Samples*
CREMA-D	7,442
TESS	2,400
RAVDESS	1,056
Total	**10,898**

Emotion Distribution in Combined Dataset

Emotion	*Number of Samples*
Angry	1,863
Disgust	1,863
Fear	1,863
Happy	1,863
Sad	1,863
Neutral	1,583
Total	**10,898**

The combined dataset was subsequently divided into training, validation, and testing subsets for model development and evaluation.

Dataset Split	*Number of Samples*
Training	8,717
Validation	1,090
Testing	1,091
Total	**10,898**

DATA PREPROCESSING

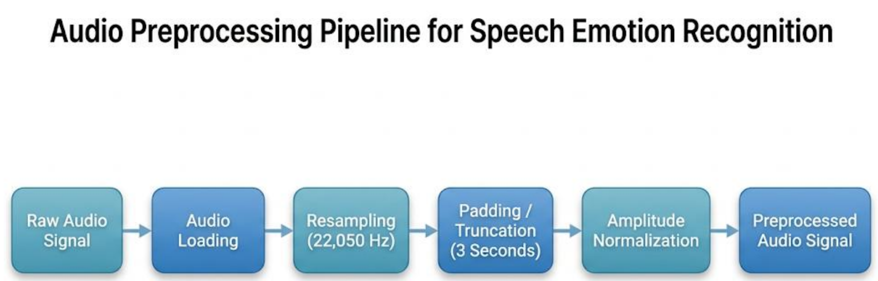
Raw speech recordings collected from different datasets often exhibit variations in sampling rate, duration, amplitude, and recording conditions. Such inconsistencies can negatively affect feature extraction and model performance. Therefore, audio preprocessing is a crucial step in the Speech Emotion Recognition pipeline, as it ensures that all speech samples are transformed into a standardized format before feature extraction.

The preprocessing pipeline adopted in the proposed system is illustrated in Figure . Initially, each audio recording was loaded and converted into a uniform format for further processing. Since the datasets used in this work were collected under different recording conditions, the audio signals were first resampled to a sampling frequency of 22,050 Hz. Resampling ensures consistency across all speech samples and provides a common temporal resolution for subsequent processing.

To maintain a fixed input size for feature extraction and deep learning model training, each speech signal was standardized to a duration of 3 seconds. Audio recordings shorter than the required duration were padded with zeros, while longer recordings were truncated. This step ensures uniform feature dimensions across the entire dataset and simplifies batch processing during training.

Amplitude normalization was then applied to reduce variations caused by differences in recording volume and microphone sensitivity. Normalization scales the audio signals to a consistent amplitude range, enabling the model to focus on emotional characteristics rather than signal intensity differences.

After completing these preprocessing steps, the resulting speech signals were standardized and ready for feature extraction. The preprocessed audio data were subsequently used to compute MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features, which served as inputs to the proposed Speech Emotion Recognition model.



DATA AUGMENTATION

Data augmentation is an important technique used to increase the diversity of training data and improve the generalization capability of deep learning models. In Speech Emotion Recognition, the availability of labeled emotional speech data is often limited. Therefore, data augmentation helps generate additional variations of speech signals without altering their underlying emotional content.

In the proposed system, data augmentation was applied only to the training dataset, while the validation and testing datasets were kept unchanged to ensure unbiased model evaluation. The objective of augmentation was to expose the model to different speech variations and improve its robustness under real-world conditions.

Three augmentation techniques were employed in this work:

1. Noise Injection

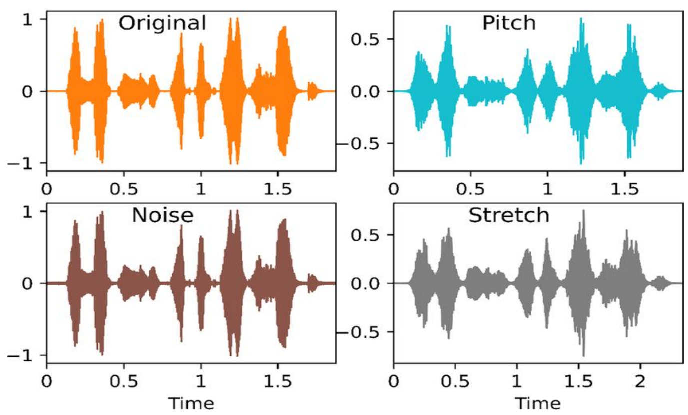
Random noise was added to the original speech signals to simulate noisy recording environments. This technique helps the model learn robust emotional representations and improves its performance when processing speech recorded in the presence of background noise.

2. Pitch Shifting

Pitch shifting modifies the pitch of a speech signal without changing its duration. Since speakers naturally differ in vocal pitch due to factors such as age and gender, this augmentation technique helps the model become less sensitive to speaker-specific variations while preserving emotional characteristics.

3. Time Stretching

Time stretching changes the speed of a speech signal without affecting its pitch. This introduces variations in speaking rate and allows the model to learn emotional patterns from speech delivered at different speeds.



Data Augmentation Techniques Applied to Speech Signals

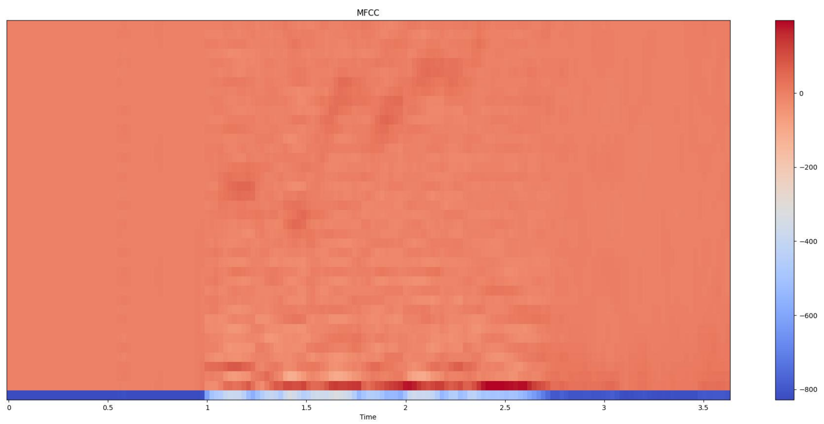
FEATURE EXTRACTION

Feature extraction is a crucial step in a Speech Emotion Recognition (SER) system, as it converts raw speech signals into meaningful numerical representations. In this work, multiple acoustic features, namely MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrograms, were extracted from each speech signal. These features capture complementary temporal and spectral characteristics of speech and serve as inputs to the proposed feature fusion model for emotion classification.

1. Mel-Frequency Cepstral Coefficients (MFCC)

Mel-Frequency Cepstral Coefficients (MFCCs) are among the most widely used features in speech processing applications. MFCCs represent the short-term power spectrum of speech signals on the Mel scale, which approximates human auditory perception. These features effectively capture the spectral characteristics of speech and provide valuable information for emotion recognition.

In this work, 40 MFCC coefficients were extracted from each speech sample. The resulting MFCC feature representation had a dimension of 40 × 130.



MFCC Representation Extracted from a Speech Signal

2. Delta MFCC

Delta MFCC features represent the first-order temporal derivatives of MFCC coefficients. They capture the rate of change of spectral information over time and provide dynamic characteristics of speech signals. These features help the model learn temporal variations associated with different emotional states.

In this work, Delta MFCC features were computed from the extracted MFCC coefficients, resulting in a feature representation of size 40 × 130.

3. Delta-Delta MFCC

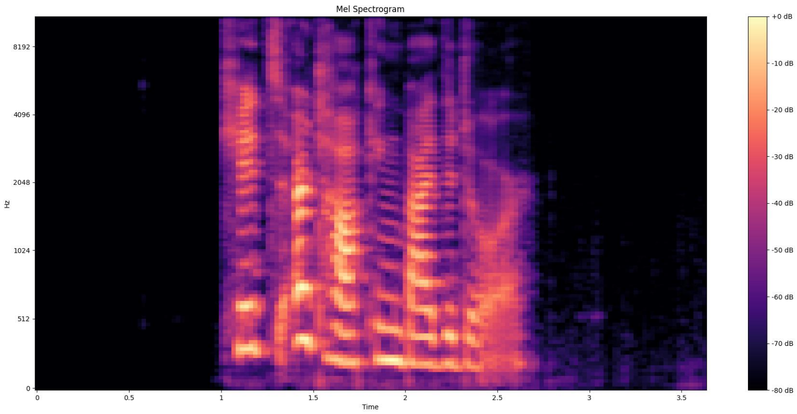
Delta-Delta MFCC features represent the second-order temporal derivatives of MFCC coefficients. These features capture acceleration information and provide additional temporal dynamics present in speech signals. The inclusion of Delta-Delta features enhances the model's ability to distinguish subtle emotional variations.

The extracted Delta-Delta MFCC feature representation also had a dimension of 40 × 130.

4. Mel Spectrogram

A Mel Spectrogram is a time-frequency representation of a speech signal obtained by mapping frequencies onto the Mel scale. Unlike MFCCs, which provide a compact representation of speech, Mel Spectrograms preserve detailed spectral information and are particularly suitable for deep learning models such as Convolutional Neural Networks (CNNs).

In this work, Mel Spectrogram features of dimension 128 × 130 were extracted from each speech sample. These features were used as input to the CNN branch of the proposed



architecture.

Mel Spectrogram Representation of a Speech Signal

Extracted Feature Representation

The extracted MFCC, Delta MFCC, and Delta-Delta MFCC features were combined to form a feature matrix of size 120 × 130. This feature representation was used as input to the BiLSTM-Attention branch of the proposed model. The Mel Spectrogram representation of size 128 × 130 was simultaneously provided to the CNN branch.

By combining multiple acoustic representations, the proposed system captures complementary temporal and spectral information, thereby improving emotion recognition performance.

FEATURE FUSION MODEL

Feature fusion is a technique used to combine information obtained from multiple feature representations in order to improve the performance of machine learning and deep learning models. In Speech Emotion Recognition, different acoustic features capture different characteristics of speech signals. Therefore, combining multiple complementary features can provide a richer representation of emotional information.

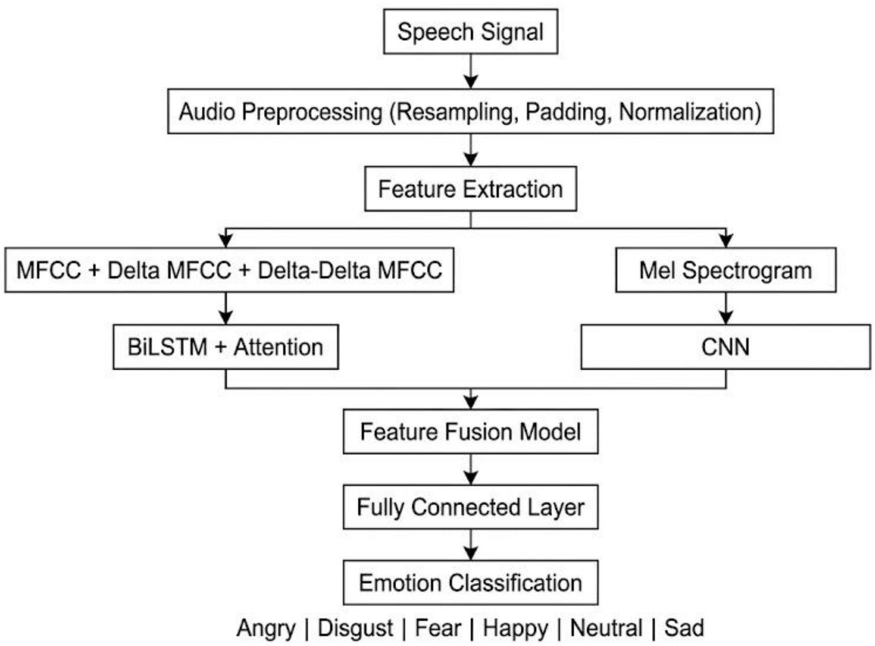
In the proposed system, four acoustic features were extracted from each speech signal: MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram. The MFCC-based features capture spectral and temporal variations in speech, while the Mel Spectrogram preserves detailed time-frequency information.

The extracted MFCC, Delta MFCC, and Delta-Delta MFCC features, each of dimension 40 × 130, were combined along the feature axis to form a unified feature matrix of size 120 × 130. This combined representation was used as input to the BiLSTM-Attention branch of the proposed model.

Simultaneously, Mel Spectrogram features of dimension 128 × 130 were provided to a separate CNN branch. The CNN branch learns spatial representations from the time-frequency structure of speech signals, while the BiLSTM-Attention branch captures temporal dependencies and contextual information.

The high-level representations learned from both branches were fused using a feature fusion layer. This fused representation combines complementary information extracted from different feature domains and serves as the final input to the emotion classification network.

The proposed feature fusion strategy enables the model to effectively utilize both spectral and temporal characteristics of speech, thereby improving emotion recognition performance and model robustness.



NEURAL NETWORK ARCHITECTURE

The proposed Speech Emotion Recognition system employs a hybrid deep learning architecture consisting of Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and an Attention mechanism. The architecture is designed to effectively capture both spectral and temporal characteristics of speech signals and improve emotion classification performance.

The model utilizes two parallel processing branches. The first branch processes Mel Spectrogram features using a CNN, while the second branch processes the combined MFCC feature matrix using BiLSTM and Attention layers. The high-level representations extracted from both branches are fused and subsequently used for emotion classification.

1. CNN Branch

The CNN branch receives Mel Spectrogram features of dimension 128 × 130 × 1 as input. Multiple convolutional layers are employed to learn spatial patterns present in the time-frequency representation of speech signals. Each convolutional layer is followed by Batch Normalization, ReLU activation, and Max Pooling operations to improve feature learning and reduce computational complexity.
The CNN branch gradually extracts high-level spectral representations that are useful for distinguishing different emotional states.

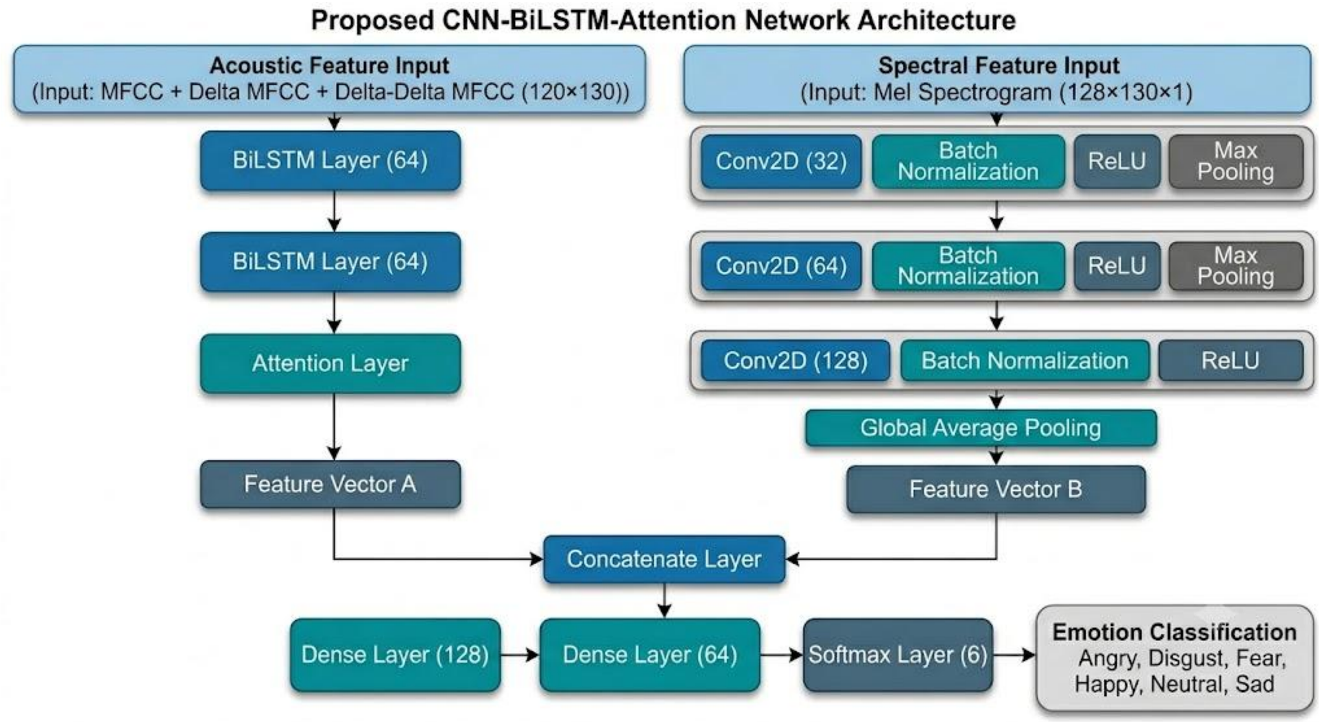
2. BiLSTM-Attention Branch

The BiLSTM branch receives the combined MFCC feature matrix of dimension 120 × 130. Bidirectional Long Short-Term Memory networks are used to capture temporal dependencies present in speech signals by processing information in both forward and backward directions.
An Attention layer is incorporated after the BiLSTM layers to enable the model to focus on emotionally relevant regions of the speech sequence. This mechanism improves the model's ability to capture important contextual information associated with different emotions.

3. Feature Fusion and Classification

The feature representations extracted by the CNN and BiLSTM-Attention branches are combined through a feature fusion layer. The fused representation is passed through fully connected layers to learn discriminative emotional representations.Finally, a Softmax output layer performs multi-class classification and predicts one of the six emotion categories: Angry, Disgust, Fear, Happy, Neutral, and Sad.

The hybrid CNN-BiLSTM-Attention architecture enables effective utilization of both spectral and temporal information, resulting in improved emotion recognition capability and model robustness.



Summary of the Proposed CNN-BiLSTM-Attention Architecture

Component	Configuration	Purpose
Acoustic Features	MFCC + Delta MFCC + Delta-Delta MFCC (120 × 130)	Capture temporal and spectral speech information
Spectral Features	Mel Spectrogram (128 × 130 × 1)	Preserve time-frequency characteristics
BiLSTM Network	2 Layers (64 Units Each)	Learn temporal dependencies in speech
Attention Layer	Applied after BiLSTM	Focus on emotionally relevant information
CNN Network	Conv2D (32, 64, 128 Filters)	Learn spatial patterns from spectrograms
Feature Fusion	Concatenation Layer	Combine features from both branches
Classification Layers	Dense (128), Dense (64), Softmax (6)	Perform emotion classification

MODEL TRAINING METHODOLOGY

The proposed Speech Emotion Recognition model was trained using a supervised deep learning approach. After feature extraction and feature fusion, the generated feature representations were provided as inputs to the CNN-BiLSTM-Attention network for emotion classification. The training process incorporated various optimization, regularization, and monitoring techniques to improve convergence and generalization performance.

Dataset Splitting

The combined dataset was divided into training, validation, and testing sets. The training set was used for model learning, the validation set was used to monitor model performance during training, and the test set was reserved for final evaluation.

DATASET SPLIT	NUMBER OF SAMPLES
TRAINING SET	8,718
VALIDATION SET	1,090
TEST SET	1,090

Training Data Augmentation

To increase data diversity and improve model robustness, data augmentation was applied only to the training set. Three augmentation techniques were employed:

- Noise Injection
- Pitch Shift
- Time Stretch

These augmented samples were combined with the original training data, resulting in a larger and more diverse dataset for model training.

Training Configuration

The model was trained using the Adam optimizer and Categorical Crossentropy loss function. Various hyperparameters were selected experimentally to achieve stable learning and effective convergence.

PARAMETER	VALUE
NUMBER OF CLASSES	6
BATCH SIZE	64
EPOCHS	50

LEARNING RATE	0.001
OPTIMIZER	Adam
LOSS FUNCTION	Categorical Crossentropy
LSTM DROPOUT	0.3
FUSION DROPOUT	0.3
CNN REGULARIZATION	L2 (1×10^{-4})
CLASS WEIGHTS	Applied

Optimization and Regularization

Several optimization and regularization techniques were employed to improve model performance and reduce overfitting. Dropout layers were incorporated within the BiLSTM and fusion layers to improve generalization. In addition, L2 regularization was applied to the convolutional layers of the CNN branch to prevent excessive weight growth and improve learning stability. Class weights were also utilized during training to address class imbalance and ensure fair learning across all emotion categories.

Training Callbacks

Multiple callback mechanisms were employed to enhance training efficiency and automatically save the best-performing model. Early Stopping was used to terminate training when validation performance stopped improving, thereby preventing unnecessary training and reducing overfitting. ReduceLROnPlateau dynamically adjusted the learning rate whenever the validation loss plateaued. Model Checkpointing was used to automatically save the model with the highest validation accuracy.

Training Callback Configuration

CALLBACK	CONFIGURATION
EARLY STOPPING	Patience = 10
REDUCELRONPLATEAU	Patience = 3
LEARNING RATE REDUCTION FACTOR	0.5
MINIMUM LEARNING RATE	1×10^{-6}
MODEL CHECKPOINT	Best Validation Accuracy

IMPLEMENTATION DETAILS

The proposed Speech Emotion Recognition system was implemented using Python and various open-source libraries for audio processing, feature extraction, machine learning, and deep learning. TensorFlow and Keras were used for model development and training, while Librosa was employed for speech signal processing and feature extraction. Scikit-Learn was used for data preprocessing and performance evaluation.

Software Environment Used for Implementation

Software / Library	Version
Python	3.12.6
TensorFlow	2.21.0
Keras	3.14.1
Librosa	0.10.2.post1
NumPy	1.26.4
Pandas	2.2.2
Scikit-Learn	1.5.1
Matplotlib	3.9.0
OpenCV	4.10.0.84

Hardware Configuration

The experiments were performed on a personal laptop. The hardware specifications used during model development and training are summarized below.

Hardware Configuration

Component	Specification
Operating System	Windows 11 Home Single Language
Processor	Intel Core i7-1255U
RAM	16 GB
Graphics	Intel Iris Xe Graphics
Development Environment	Visual Studio Code

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The performance of the proposed CNN-BiLSTM-Attention based Speech Emotion Recognition system was evaluated using the test dataset containing six emotion classes: Angry, Disgust, Fear, Happy, Neutral, and Sad. The effectiveness of the model was assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Training and Test Performance

The proposed model was trained for a maximum of 50 epochs using the Adam optimizer and Categorical Crossentropy loss function. During training, Early Stopping, ReduceLROnPlateau, and Model Checkpoint callbacks were employed to improve convergence and prevent overfitting.

The model demonstrated effective learning during the training phase and achieved approximately **85.3% training accuracy**. The highest validation accuracy obtained during training was **70.61%**, while the final test accuracy achieved on unseen speech samples was **71.82%**.

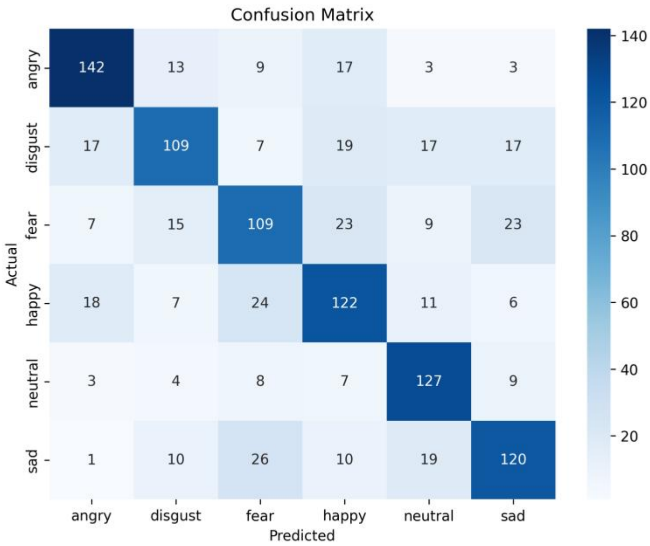
These results indicate that the model successfully learned meaningful emotional representations from speech signals and was able to generalize reasonably well to unseen data.

Overall Performance Metrics

Metric	Value
Training Accuracy	85.30%
Best Validation Accuracy	70.61%
Test Accuracy	71.82%
Macro Precision	72.90%
Macro Recall	73.15%
Macro F1-Score	72.86%
Weighted F1-Score	71.68%

Confusion Matrix Analysis

A confusion matrix was generated to visualize the classification performance of the proposed model across all six emotion categories.



The confusion matrix shows that the model achieved strong recognition performance for **Angry**, **Happy**, and **Neutral** emotions. Among all classes, Neutral emotion exhibited the highest recognition rate, indicating that the model was able to effectively capture its acoustic characteristics.

Some confusion was observed between Fear and Sad emotions, as well as between Disgust and other emotion categories. Such misclassifications are expected because several emotions share similar acoustic and prosodic patterns, making them difficult to distinguish.

Overall, the confusion matrix demonstrates that the proposed feature fusion framework was capable of learning discriminative emotional representations from speech signals.

Classification Performance for Individual Emotions

Emotion	Precision (%)	Recall (%)	F1-Score (%)
Angry	75.53	75.94	75.73
Disgust	68.99	59.60	63.37
Fear	60.56	61.60	64.08
Happy	64.62	68.89	65.12
Neutral	68.28	80.38	73.84
Sad	67.42	64.52	65.93

Among all emotion categories, **Angry** achieved the highest F1-score (75.73%), while **Neutral** achieved the highest recall (80.38%). The comparatively lower performance for Fear and Disgust emotions indicates the challenge of distinguishing emotions with overlapping speech characteristics.

Performance Summary

The experimental results demonstrate that the integration of MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features with the proposed CNN-BiLSTM-Attention architecture effectively captured both spectral and temporal information present in speech signals. The model achieved satisfactory performance across six emotion classes and successfully learned meaningful emotional representations from speech data.

DISCUSSION

The experimental results indicate that the proposed Speech Emotion Recognition system was able to effectively learn emotional patterns from speech signals using the combination of feature fusion and deep learning techniques. The integration of MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features provided complementary information, enabling the model to capture both temporal and spectral characteristics of speech.

The hybrid CNN-BiLSTM-Attention architecture contributed significantly to the overall performance of the system. The CNN branch effectively extracted spatial patterns from Mel Spectrogram features, while the BiLSTM and Attention mechanism captured temporal dependencies and emphasized emotionally relevant information. The feature fusion strategy further improved the representation learning capability of the model.

The model achieved a training accuracy of approximately 85.3%, a best validation accuracy of 70.61%, and a test accuracy of 71.82%. The close agreement between validation and test accuracy indicates that the model was able to generalize reasonably well to unseen speech samples. Analysis of the confusion matrix revealed strong performance for Angry and Neutral emotions, whereas comparatively lower performance was observed for Fear and Disgust emotions due to similarities in their acoustic characteristics.

Overall, the obtained results demonstrate that the proposed feature fusion approach combined with the CNN-BiLSTM-Attention architecture provides an effective framework for Speech Emotion Recognition and is capable of achieving satisfactory emotion classification performance across multiple emotion categories.

LIMITATIONS OF THE SYSTEM

- The model was trained using publicly available datasets, which may not fully represent real-world speech variability and recording conditions.
- Emotional expression varies significantly among different speakers, making emotion recognition a challenging task.
- Background noise, microphone quality, and environmental conditions can affect the performance of the system.
- Certain emotions such as Fear, Disgust, and Sad exhibit similar acoustic characteristics, which may lead to misclassification.
- The current system is limited to six predefined emotion categories and does not consider mixed or continuous emotional states.
- The performance of deep learning models can be further improved using larger and more diverse emotional speech datasets.

FUTURE WORK

- Integration of pre-trained speech models such as **Wav2Vec 2.0**, **HuBERT**, and **WavLM** for improved feature representation and classification accuracy.
- Utilization of larger and more diverse emotional speech datasets to improve model generalization.
- Development of **multilingual emotion recognition systems** capable of handling different languages, accents, and speaking styles.
- Implementation of **real-time emotion recognition** using live microphone input.
- Exploration of **Transformer-based architectures** and advanced attention mechanisms for better emotion representation learning.
- Improvement of robustness against background noise and real-world recording conditions.
- Deployment of the system as a **web-based or mobile application** for practical real-world use.
- Extension of the framework to support a larger number of emotion categories and mixed emotional states.

CONCLUSION

Speech Emotion Recognition is an important research area that enables intelligent systems to understand and interpret human emotions from speech signals. The ability to automatically recognize emotions has significant applications in human-computer interaction, healthcare, education, customer support systems, virtual assistants, and various emotion-aware technologies.

In this project, a deep learning-based Speech Emotion Recognition system was developed to classify six different emotional states from speech signals. A feature fusion approach was adopted by combining MFCC, Delta MFCC, Delta-Delta MFCC, and Mel Spectrogram features to capture both temporal and spectral characteristics of speech. The extracted features were then processed using a hybrid CNN-BiLSTM-Attention architecture designed to learn complex emotional patterns from speech data.

The CNN branch effectively extracted high-level spatial features from Mel Spectrogram representations, while the BiLSTM and Attention mechanism captured temporal dependencies and emphasized emotionally relevant information from MFCC-based features. The learned representations from both branches were fused to improve emotion discrimination and classification performance.

The model was trained and evaluated on a combined dataset consisting of RAVDESS, CREMA-D, and TESS speech samples. Experimental evaluation demonstrated satisfactory performance across six emotion categories, achieving a test accuracy of 71.82% along with balanced precision, recall, and F1-score values. Analysis of the confusion matrix and classification report indicated that the proposed system was able to effectively recognize emotions such as Angry and Neutral while maintaining reasonable performance across other emotional categories.

Overall, the results obtained in this work demonstrate that the combination of feature fusion and deep learning provides an effective framework for Speech Emotion Recognition. The proposed system successfully learned meaningful emotional representations from speech signals and showed the capability to generalize to unseen data. The work carried out during this internship provides a strong foundation for future research involving advanced speech representation models, multilingual emotion recognition, and real-time emotion-aware intelligent systems.

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CERTIFICATE

This is to certify that **Mr. Arsh Mohan Nishant**, a Bachelor of Technology (B.Tech) student in **Computer Science and Engineering at KIIT University, Bhubaneswar**, has successfully completed the project titled **“Speech Emotion Recognition Using Deep Learning”** at **Defence Electronics Applications Laboratory (DEAL), Dehradun**, as a part of his internship.

The internship was carried out during the period from **12th May 2026 to 26th June 2026**, under the guidance and supervision of **Mr. Sandeep Kishore,** Scientist ‘E’, Artificial Intelligence & Advanced Tactical Networking and Communication Systems (AI & ATNCS), DEAL, DRDO, Dehradun, Uttarakhand.

During the internship, Mr. Arsh Mohan Nishant demonstrated dedication, sincerity, and strong technical skills in the field of Artificial Intelligence and Deep Learning. He successfully worked on the development of a Speech Emotion Recognition system involving audio preprocessing, feature extraction, deep learning model development, training, and performance evaluation.

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